Composition Operators between Sobolev Spaces

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Historical notes

Main result

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Motivating question

The simplest change-of-variables problem: I

Let AC^p, $1 \leqslant p < \infty$, be the semi-normed space of absolutely continuous functions u such that $u' \in L^p$, $\|u\|_{AC^p} = \|u'\|_{L^p}$. Let $\varphi\colon [a,b] \to [c,d]$ be a homeomorphism.

We consider the change-of-variables problem on AC^p. Suppose that

$$\varphi^* u := u \circ \varphi \in AC^p[a,b] \quad \text{for every} \quad u \in AC^p[c,d],$$

and there exists a constant $K > 0$ such that the inequality

$$\|\varphi^* u\|_{AC^p[a,b]} \leqslant K \|u\|_{AC^p[c,d]} \quad \text{holds for every} \quad u \in AC^p.$$

We say that φ^* is a composition operator.

The simplest change-of-variables problem: II

Let AC^p, $1 \leqslant p < \infty$, be the semi-normed space of absolutely continuous functions u such that $u' \in L^p$, $\|u\|_{AC^p} = \|u'\|_{L^p}$. Let $\varphi\colon [a, b] \to [c, d]$ be a homeomorphism. Suppose that φ^* is a bounded composition operator on AC^p.

Question

What geometric restrictions on φ follow from the boundedness of the change-of-variables operator? Are they sufficient?

Answer

For $p > 1$, φ must be Lipschitz. Determine $\|\varphi^*\|$.

For $p = 1$, every absolutely continuous φ induces a bounded composition operator. Determine $\|\varphi^*\|$.

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Historical notes

Sobolev's conjecture

- In 1941, S. L. Sobolev proved that C^1 quasi-isometries preserve the $L^{1,p}$ spaces in $\mathbb{R}^n$, $n \geq 2$.
  He conjectured that all isomorphisms of Sobolev spaces $L^{1,p}$ are quasi-isometries.
- It follows from V. G. Maz'ya's PhD thesis that C^1 isomorphisms of $L^{1,p}$ spaces are induced by
  - quasi-isometries if $p \neq n$,
  - quasiconformal mappings if $p = n$.

Reshetnyak's problem

In 1968, Yu. G. Reshetnyak posed the problem of characterizing all isomorphisms of the Sobolev space $L^{1,n}$ in $\mathbb{R}^n$ induced by quasiconformal mappings.

- In 1974, E. A. Poletsky answered this question for the space $L^{1,n} \cap C^1$.
- In 1975, S. K. Vodopyanov and V. M. Gol'dshtejn gave a full solution of Reshetnyak's problem.

In the 1980s, the study of mappings that induce bounded composition operators on Sobolev spaces began.

Composition operators: I

Characterizations of homeomorphisms that induce bounded composition operators on Sobolev spaces have been established in several model settings:

- on Carnot groups (including Euclidean space), see S. K. Vodopyanov and A. D. Ukhlov (1998), S. K. Vodopyanov and N. A. Evseev (2022);
- on Riemannian manifolds, see S. K. Vodopyanov (2024 and 2025).

The list of publications is not exhaustive; additional work includes results on measurable mappings and related topics.

Composition operators on metric measure spaces

Related results include:

- M. Mocanu (2006) proved that quasiconformal (QC) mappings on certain metric measure spaces (m.m.s.) induce continuous composition operators on Newtonian spaces $N_0^{1,Q}$;
- QC mappings induce bounded composition operators on Triebel–Lizorkin spaces, see P. Koskela, D. Yang, Y. Zhou (2011), and on Bessel potential spaces, see M. Olivia, M. Prats (2017);
- Quasisymmetric mappings induce isomorphisms between Hajłasz–Triebel–Lizorkin spaces, see M. Bourdon, H. Pajot (2003);

The list of publications is not exhaustive; additional work includes results on composition operators on other functional spaces and related topics.

Main result

Characterization of bounded composition operators

Theorem

Let (X, d, μ), (Y, ρ, ν) be metric measure spaces, and let $\varphi\colon X \to Y$ be a homeomorphism. Fix $1 \leqslant q \leqslant p < \infty$.

The composition operator $\varphi^*\colon D^{1,p}(Y) \cap LIP(Y) \to D^{1,q}(X)$ is bounded if and only if

1. $\varphi \in D^{1,q}_{\mathrm{loc}}(X; Y)$,
2. φ *has finite distortion, i.e.,* $|D\varphi| = 0$ *a.e. on the set* $Z_\varphi = \{x \mid \mathcal{J}\varphi(x) = 0\}$,
3. *the operator distortion function* $K_{q,p}(\cdot, \varphi)$ *belongs to* $L^{\varkappa}(X)$, *where* $1/\varkappa = 1/q - 1/p$.

Moreover, $\|\varphi^*\| = \|K_{q,p}(\cdot, \varphi)\|_{L^{\varkappa}(X)}$.

This theorem was proven for $q = p$ by Sboev (2025, Sib. Math. J.), for $q < p$ by Vodopyanov, Sboev (2025, Rus. Math. Surveys; 2026, Mat. Sb.).

Aim of the talk

Here and elsewhere, Ω and Ω' are domains in $\mathbb{R}^n$.

Theorem (Case of Euclidean spaces; $q < p$)

Let $\varphi\colon \Omega \to \Omega'$ be a homeomorphism. Fix $1 \leqslant q < p < \infty$. The composition operator $\varphi^*\colon L^{1,p}(\Omega') \cap LIP(\Omega') \to L^{1,q}(\Omega)$ is bounded if and only if

1. *$\varphi \in L^{1,q}_{\mathrm{loc}}(\Omega;\Omega')$,*
2. *φ has finite distortion, i.e., $D\varphi = 0$ a.e. on the set $Z_\varphi = \{x \mid \mathcal{J}\varphi(x) = 0\}$,*
3. *the operator distortion function $K_p(\cdot,\varphi)$ belongs to $L^{\varkappa}(\Omega)$, where $1/\varkappa = 1/q - 1/p$.*

Moreover, $\|\varphi^*\| = \|K_p(\cdot,\varphi)\|_{L^{\varkappa}(\Omega)}$.

Preliminaries

Preliminaries: Lipschitz functions

Let Ω be a domain in $\mathbb{R}^n$, $n \geq 2$.

- A function $u\colon \Omega \to \mathbb{R}$ is called Lipschitz continuous, if $|u(x) - u(y)| \leqslant L|x - y|$ for all x, $y \in \Omega$.
  The smallest constant L above is denoted by $\mathrm{Lip}(u)$.
- The space of all Lipschitz functions is denoted by $LIP(\Omega)$.

Preliminaries: Sobolev functions

Let Ω be a domain in $\mathbb{R}^n$, $n \geq 2$. Fix $1 \leqslant p < \infty$.

- A function $u\colon \Omega \to \mathbb{R}$ belongs to the space $L^{1,p}(\Omega)$, if $u \in L^1_{\mathrm{loc}}(\Omega)$ and $\nabla u \in L^p(\Omega)$.
  Here ∇u is the generalized gradient of u that is

$$\int_\Omega u \operatorname{div} \theta \, dx = -\int_\Omega (\nabla u, \theta) \, dx \quad \text{for all} \quad \theta \in C_0^1(\mathbb{R}^n; \mathbb{R}^n).$$

Preliminaries: Truncations

- For $u,\ v \in L^{1,p}$, the following holds a.e.:

$$|\nabla \max(u, v)| = |\nabla u|\chi_{\{u \geq v\}} + |\nabla v|\chi_{\{v > u\}}.$$

The equality for $|\nabla \min(u, v)|$ is analogous.

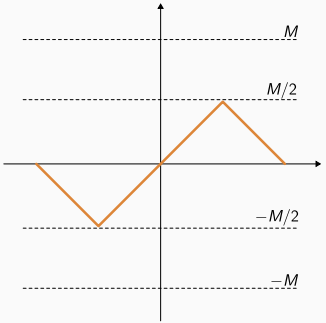


Fig.: Function R_M.

From this we obtain that $|\nabla(R_M \circ u)| = |\nabla u|$ a.e. for every $u \in L^{1,p}$, $|u| \leqslant M$.

Preliminaries: Sobolev mappings

- We say that a measurable mapping $\varphi\colon \Omega \to \Omega'$ belongs to the Sobolev space $L^{1,p}(\Omega;\Omega')$, $1 \leqslant p < \infty$, if every function $\varphi_i \in L^{1,p}(\Omega)$, $i = 1, \dots, n$.

This is equivalent to the boundedness of composition operator φ^*

$$LIP(\Omega') \xrightarrow{\varphi^*} L^{1,p}(\Omega)$$

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Preliminaries: Mappings of finite distortion

Definition

A mapping $\varphi \in W^{1,p}_{\mathrm{loc}}(\Omega;\Omega')$, $1 \leqslant p < \infty$, has finite distortion, if

$$|D\varphi| = 0 \quad \text{a.e. on the set} \quad Z_\varphi = \{x \mid \mathcal{J}\varphi(x) = 0\}.$$

Let us define an operator distortion function

$$K_p(x,\varphi) = \begin{cases} \dfrac{|D\varphi|(x)}{\mathcal{J}\varphi(x)^{\frac{1}{p}}}, & \text{if } \mathcal{J}\varphi(x) \neq 0, \\ 0, & \text{elsewhere.} \end{cases}$$

Hence one can consider mappings with bounded distortion, integrable distortion and so on.

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Hence one can consider mappings with bounded distortion, integrable distortion and so on.

Preliminaries: Examples of mappings

Example

Let $\varphi\colon D \to \varphi(D)$ be a C^1-diffeomorphism. Suppose that $\Omega \Subset D$.

Then $K_p(\cdot, \varphi)$ is bounded on Ω and hence integrable with any power.

Example

Let $\varphi\colon \mathbb{R}^n \setminus \{0\} \to \mathbb{R}^n \setminus \{0\}$ be a radial stretching:

$$\varphi(x) = |x|^{\alpha-1}x.$$

This is a quasiconformal mapping ($K_n(\cdot, \varphi) \leqslant K$) and hence on any subdomain of finite measure $K_n(\cdot, \varphi)$ is integrable in any power.

For $p \neq n$ we have

$$K_p(x, \varphi) = c_\alpha |x|^{(\alpha-1)(1-n/p)}.$$

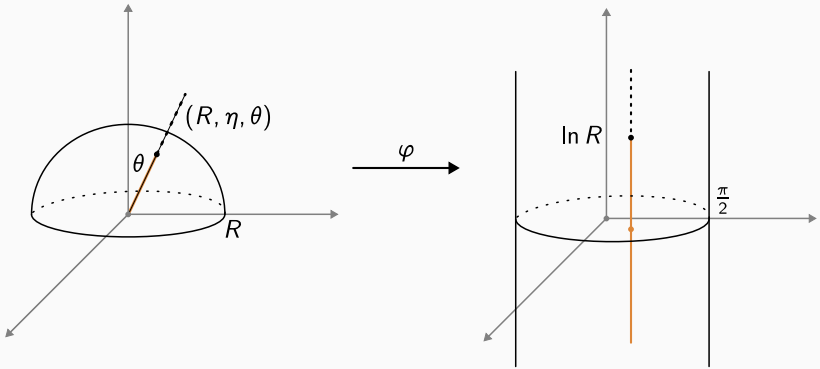
Example

Let $\varphi\colon \mathbb{R}^3_+ \to \varphi(\mathbb{R}^3_+)$ be a mapping between spherical and cylindrical coordinates

$$\varphi(r, \eta, \theta) = (\theta, \eta, \ln r).$$

This is a quasiconformal mapping and hence on any subdomain of finite measure $K_n(\cdot, \varphi)$ is integrable in any power.

Preliminaries: Examples of mappings



Measure associated with a composition operator

How to find the properties of φ?

Let $\varphi\colon \Omega \to \Omega'$ be a homeomorphism, and suppose that φ^* is bounded. Hence, for every $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$, $\varphi^* u = u \circ \varphi \in L^{1,q}(\Omega)$ and

$$\|\varphi^* u\|_{L^{1,q}(\Omega)} \leqslant \|\varphi^*\| \|u\|_{L^{1,p}(\Omega')}.$$

Question

How can we extract geometric properties of φ?

Our answer

We will construct a measure that encodes geometric properties of φ.

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Operator norms

Suppose that $\varphi^*\colon L^{1,p}(\Omega') \cap LIP(\Omega') \to L^{1,q}(\Omega)$, $1 \leqslant q < p < \infty$ is bounded.

What are quantitative characteristics of φ^*?

- Norm $\|\varphi^*\|$.
- If U is open, we can define "norm"

$$\|\varphi^*_U\| = \sup \left(\int\limits_{\varphi^{-1}(U)} |\nabla \varphi^* u|^q \, dx \right)^{\frac{1}{q}}$$

where supremum is taken over all $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$ such that $\|u\|_{L^{1,p}(U)} = 1$.

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Set function

Let $1/\varkappa = 1/q - 1/p$ (equiv., $q/\varkappa + q/p = 1$).

For every Borel set $B \subset \Omega'$ we define

$$\Psi(B) = \sup \left(\int\limits_{\varphi^{-1}(B)} |\nabla \varphi^* u|^q \, dx \right)^{\frac{\varkappa}{q}},$$

where supremum is taken over all $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$ such that $\|u\|_{L^{1,p}(B)} = 1$.

From the definition, one obtains that

$$\int\limits_{\varphi^{-1}(B)} |\nabla \varphi^* u|^q \, dx \leqslant \Psi(B)^{\frac{q}{\varkappa}} \left(\int\limits_B |\nabla u|^p \, dy \right)^{\frac{q}{p}}$$

for every $B \in \mathscr{B}(\Omega')$ and all $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$.

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for every $B \in \mathscr{B}(\Omega')$ and all $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$.

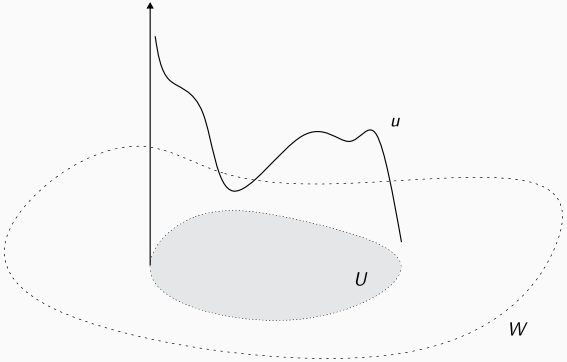
Monotonicity

Suppose that $U \subset W$. Our goal is to prove that

$$\int\limits_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx \leqslant \Psi(W)^{\frac{q}{\varkappa}} \left(\int\limits_U |\nabla u|^p \, dy \right)^{\frac{q}{p}}$$

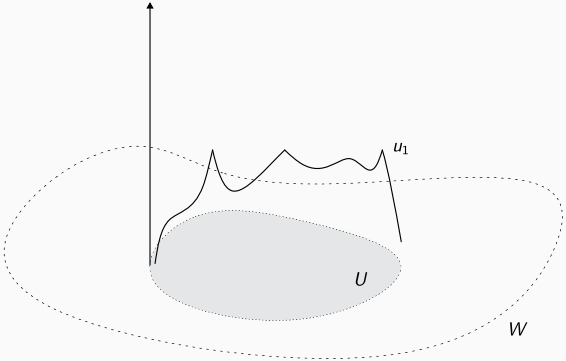
for every $u \in L^{1,p}(\Omega') \cap LIP(\Omega')$.

Monotonicity



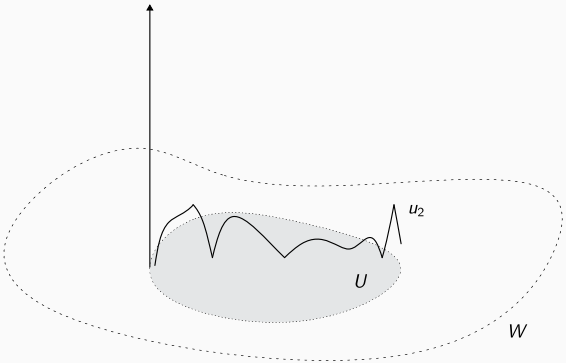
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx$$

Monotonicity



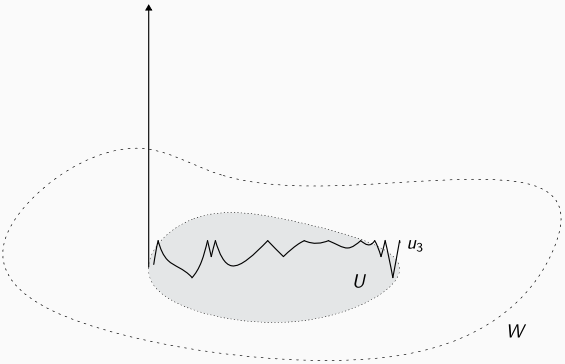
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(u_1 \circ \varphi)|^q \, dx$$

Monotonicity



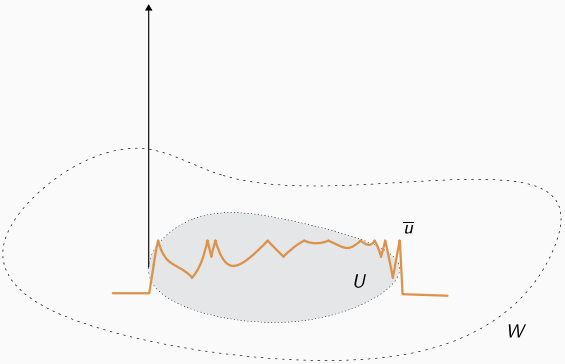
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(u_1 \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(u_2 \circ \varphi)|^q \, dx$$

Monotonicity



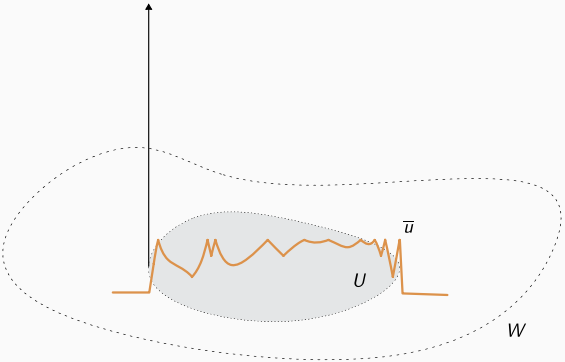
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(u_1 \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(u_2 \circ \varphi)|^q \, dx = \dots$$

Monotonicity



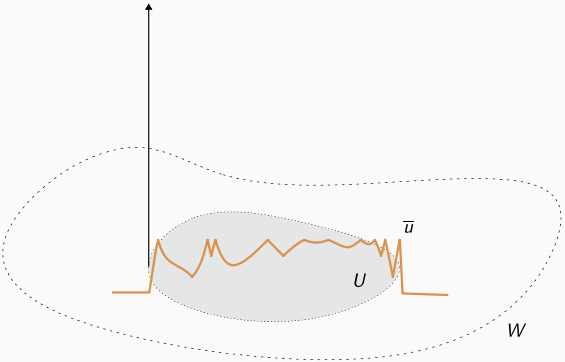
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx = \int_{\varphi^{-1}(U)} |\nabla(\overline{u} \circ \varphi)|^q \, dx \approx \int_{\varphi^{-1}(W)} |\nabla(\overline{u} \circ \varphi)|^q \, dx$$

Monotonicity



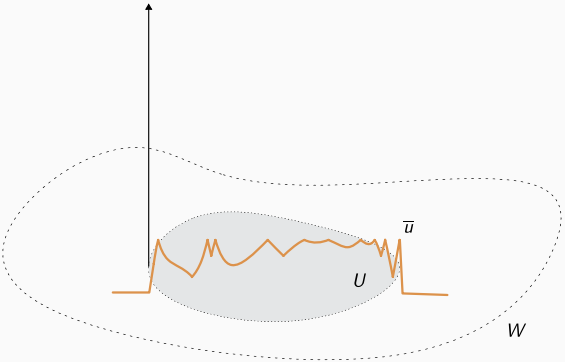
$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx \approx \int_{\varphi^{-1}(W)} |\nabla(\overline{u} \circ \varphi)|^q \, dx \leqslant \Psi(W)^{\frac{q}{\varkappa}} \left(\int_W |\nabla \overline{u}|^p \, dy \right)^{\frac{q}{p}}$$

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$$\int_{\varphi^{-1}(U)} |\nabla(u \circ \varphi)|^q \, dx \leqslant \Psi(W)^{\frac{q}{\varkappa}} \left(\int_U |\nabla u|^p \, dy \right)^{\frac{q}{p}} \quad \Longrightarrow \quad \Psi(U) \leqslant \Psi(W)$$

Measure associated with a composition operator

Theorem

Suppose that $\varphi\colon \Omega \to \Omega'$ *is a measurable mapping, and* $\varphi^*\colon L^{1,p}(\Omega') \cap LIP(\Omega') \to L^{1,q}(\Omega)$ *is bounded,* $1 \leqslant q < p < \infty$.

Then the set function Ψ *extends to a finite regular Borel measure on* Ω'

$$\widetilde{\Psi}(A) = \inf\{\Psi(U) \mid U \supset A,\ U \text{ open}\}$$

where A *is an arbitrary set in* Ω'.

We will write Ψ instead of $\widetilde{\Psi}$.

Proof of the main result

Aim of the talk

Theorem (Case of Euclidean spaces; $q < p$)

Let $\varphi\colon \Omega \to \Omega'$ be a homeomorphism. Fix $1 \leqslant q < p < \infty$. The composition operator $\varphi^*\colon L^{1,p}(\Omega') \cap LIP(\Omega') \to L^{1,q}(\Omega)$ is bounded if and only if

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Moreover, $\|\varphi^*\| = \|K_p(\cdot,\varphi)\|_{L^{\varkappa}(\Omega)}$.

Regularity

Fix a bounded open set $U \subset \Omega'$. By the definition of Ψ, we have

$$\int\limits_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q \, dx \leqslant \Psi(U)^{\frac{q}{\varkappa}} \left(\int\limits_U |\nabla u|^p \, dy \right)^{\frac{q}{p}}$$

$$\leqslant \Psi(U)^{\frac{q}{\varkappa}} \left(\int\limits_U \operatorname{Lip}(u; U)^p \, dy \right)^{\frac{q}{p}} = \Psi(U)^{\frac{q}{\varkappa}} |U|^{\frac{q}{p}} \operatorname{Lip}(u; U)^q.$$

Hence, the operator

$$\varphi_U^* \colon LIP(U) \to L^{1,q}(\varphi^{-1}(U))$$

is bounded. It follows that $\varphi \in L^{1,q}_{\mathrm{loc}}(\Omega; \Omega')$.

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Finite distortion

$$\int_{\varphi^{-1}(U)} |\nabla(\varphi^* u)|^q \, dx \leqslant \Psi(U)^{\frac{q}{\varkappa}} |U|^{\frac{q}{p}} \operatorname{Lip}(u; U)^q$$

Let $Z_\varphi = \{x \in \Omega \mid \mathcal{J}\varphi(x) = 0\}$. Then $|\varphi(Z_\varphi)| = 0$, and hence

$$\int_{Z_\varphi} |D\varphi|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} |U|^{\frac{q}{p}} = 0.$$

It follows that $|D\varphi| = 0$ a.e. on Z_φ.

Analogously, one can prove that Ψ is absolutely continuous with respect to Lebesgue measure. We denote the Radon–Nykodym derivative by Ψ'.

Finite distortion

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Analogously, one can prove that Ψ is absolutely continuous with respect to Lebesgue measure. We denote the Radon–Nykodym derivative by Ψ'.

Finite distortion

$$\int\limits_{\varphi^{-1}(U)} |D\varphi|^q \, dx \leqslant \Psi(U)^{\frac{q}{\varkappa}} |U|^{\frac{q}{p}}$$

Let $Z_\varphi = \{x \in \Omega \mid \mathcal{J}\varphi(x) = 0\}$. Then $|\varphi(Z_\varphi)| = 0$, and hence

$$\int\limits_{Z_\varphi} |D\varphi|^q \, d\mu \leqslant \Psi(U)^{\frac{q}{\varkappa}} |U|^{\frac{q}{p}} = 0.$$

It follows that $|D\varphi| = 0$ a.e. on Z_φ.

Analogously, one can prove that Ψ is absolutely continuous with respect to Lebesgue measure. We denote the Radon–Nykodym derivative by Ψ'.

Integral bounds

For simplicity, suppose that φ has the Lusin N-property. Let $B \in \mathscr{B}(\Omega)$ be a Borel subset, and put $U = \varphi(B)$. By the change-of-variables formula, we obtain

$$\int_B |D\varphi|^q \, dx \leqslant \left(\int_{\varphi(B)} \Psi' \, dy \right)^{\frac{q}{\varkappa}} \left(\int_B \mathcal{J}\varphi(x) \, dx \right)^{\frac{q}{p}}$$

$$= \left(\int_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) \, dx \right)^{\frac{q}{\varkappa}} \left(\int_B \mathcal{J}\varphi(x) \, dx \right)^{\frac{q}{p}}.$$

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$$\int_B |D\varphi|^q \, dx \leqslant \left(\int_B \Psi'(\varphi(x)) \mathcal{J}\varphi(x) \, dx \right)^{\frac{q}{\varkappa}} \left(\int_B \mathcal{J}\varphi(x) \, dx \right)^{\frac{q}{p}}, \qquad \frac{q}{\varkappa} + \frac{q}{p} = 1,$$

$$\Downarrow \ ?$$

$$|D\varphi|^q(x) \leqslant (\Psi'(\varphi(x)) \mathcal{J}\varphi(x))^{\frac{q}{\varkappa}} \mathcal{J}\varphi(x)^{\frac{q}{p}} \quad \text{a.e.}$$

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$$\Downarrow \text{ a converse to Hölder's inequality}$$

$$|D\varphi|^q(x) \leqslant (\Psi'(\varphi(x)) \mathcal{J}\varphi(x))^{\frac{q}{\varkappa}} \mathcal{J}\varphi(x)^{\frac{q}{p}} \quad \text{a.e.}$$

Pointwise estimate II

$$|D\varphi|^q(x) \leqslant (\Psi'(\varphi(x))\mathcal{J}\varphi(x))^{\frac{q}{\varkappa}}\mathcal{J}\varphi(x)^{\frac{q}{p}} \quad \text{a.e.}$$

It follows that

$$K_p(x,\varphi) = \frac{|D\varphi|(x)}{(\mathcal{J}\varphi(x))^{1/p}} \leqslant (\Psi'(\varphi(x))\mathcal{J}\varphi(x))^{\frac{1}{\varkappa}}$$

if $\mathcal{J}\varphi(x) \neq 0$. If $\mathcal{J}\varphi(x) = 0$ then by definition $K_p(x,\varphi) = 0$.

Moreover, $\|K_p(x,\varphi)\|_{L^\varkappa(\Omega)} \leqslant \Psi(\Omega')^{\frac{1}{\varkappa}} = \|\varphi^*\|$.

Aim of the talk

Theorem (Case of Euclidean spaces; $q < p$)

Let $\varphi\colon \Omega \to \Omega'$ be a homeomorphism. Fix $1 \leqslant q < p < \infty$. The composition operator $\varphi^*\colon L^{1,p}(\Omega') \cap LIP(\Omega') \to L^{1,q}(\Omega)$ is bounded if and only if

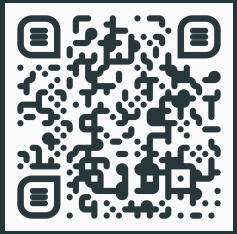
1. $\varphi \in L^{1,q}_{\mathrm{loc}}(\Omega;\Omega')$,
2. *φ has finite distortion, i.e., $D\varphi = 0$ a.e. on the set $Z_\varphi = \{x \mid \det D\varphi(x) = 0\}$,*
3. *the operator distortion function $K_p(\cdot,\varphi)$ belongs to $L^{\varkappa}(\Omega)$, where $1/\varkappa = 1/q - 1/p$.*

Moreover, $\|\varphi^*\| = \|K_p(\cdot,\varphi)\|_{L^{\varkappa}(\Omega)}$.

Composition Operators between Sobolev Spaces

Danil A. Sboev

Q&A



Slides

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